

B.Tech Computer Science & Engineering

Sample Course Syllabus (For CurricuAlign AI Backend Testing)

Course Information

Course: Artificial Intelligence Fundamentals

Semester: 6

Credits: 4

Course Objectives

Understand Artificial Intelligence, Machine Learning, Deep Learning, NLP, Computer Vision, Knowledge Graphs and modern AI deployment.

Module 1

Introduction to AI, Intelligent Agents, Search Algorithms, BFS, DFS, A*.

Module 2

Machine Learning, Classification, Regression, Decision Trees, Random Forest, SVM, KNN.

Module 3

Deep Learning, Neural Networks, CNN, RNN, LSTM, Transformers, Large Language Models (LLMs).

Module 4

Natural Language Processing, Prompt Engineering, Embeddings, Retrieval-Augmented Generation (RAG), Chatbots.

Module 5

Computer Vision, Image Processing, OpenCV, YOLO, OCR.

Module 6

MLOps, Docker, Git, GitHub, FastAPI, REST APIs, ChromaDB, Vector Databases, CI/CD.

Laboratory

Python, NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, FastAPI, Docker.

Course Outcomes

CO1: Apply AI concepts.

CO2: Develop ML models.

CO3: Build NLP applications.

CO4: Deploy AI systems.

Assessment

Mid Semester: 30 Marks
End Semester: 50 Marks
Practical: 20 Marks